Transistors, relays & MOSFETs

ENGINEERING WITH MR KELLY

Higher · Electronics & control · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how a transistor works as a switch (about 0.7 V base–emitter)
- ☐ explain the purpose of a relay and a protection diode
- ☐ calculate base current, collector current and current gain (hFE)
- ☐ describe MOSFET threshold voltage and an advantage of a MOSFET over a BJT
- ☐ use a MOSFET graph to find drain–source current

I DO — Worked example

Read how a full-mark answer is built

A BJT has a 2.2 kΩ base resistor. The input voltage is 5 V and Vbe = 0.7 V. hFE = 150.

1 Calculate the base current and collector current. (3 marks)

Full-mark answer each ✓ is one mark

VRb = 5 − 0.7 = 4.3 V ✓. Ib = 4.3 ÷ 2200 = 1.95×10^{-3} A ✓. Ic = hFE × Ib = 150 × 1.95×10^{-3} = 0.29 A ✓.

Why marks are not awarded

“Ib = 5 ÷ 2200 = 2.27 mA, Ic = 150 × 2.27 mA = 0.34 A”

1 mark (follow-on for Ic): the 0.7 V drop was missed.

WE DO — Guided practice

Use the starters and word bank to help you

An engineer replaces a BJT with a MOSFET.

2 Describe one advantage of the MOSFET over the BJT. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

resistance · switch · higher · current · currents · little · controller · almost

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A MOSFET is controlled by…
☐ Base current ☐ Gate voltage ☐ Collector current ☐ Light

b) IB = 2 mA and hFE = 150. IC is…
☐ 75 mA ☐ 300 mA ☐ 152 mA ☐ 0.3 mA

c) An NPN transistor switches on at a base voltage of about…
☐ 0.07 V ☐ 0.7 V ☐ 7 V ☐ 70 V

d) Which component lets a 5 V circuit switch a 230 V heater?
☐ LDR ☐ Relay ☐ Diode ☐ Buzzer

4 Key terms: write the letter of the matching description next to each term.

1. **Transistor switch-on voltage**	____	**A** The gate-source voltage at which the MOSFET starts to conduct.
2. **Base resistor**	____	**B** Across the relay coil; protects the transistor from reverse voltage when the coil switches off.
3. **Relay**	____	**C** Lets a low-current circuit switch a separate high-current circuit.
4. **Protection diode**	____	**D** Limits base current to protect the transistor.
5. **Current gain**	____	**E** hFE = IC ÷ IB
6. **MOSFET threshold voltage**	____	**F** About 0.7 V at the base.

A MOSFET has a threshold voltage of 2.1 V.

5 Describe what is meant by threshold voltage. (1 mark)

A transistor drives a relay coil. A diode is connected across the coil.

6 Explain the purpose of the diode. (2 marks)

A relay is used between a transistor and a mains lamp.

7 Explain why a relay is used. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?